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Perioperative Analgesia, Patient Communication, and Recovery Measurement in Breast Reconstruction Revision: A PRISMA-ScR Scoping Review and Integrative Framework

Review type	Scoping review
Reporting standard	PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews)
Methodology	JBI scoping review methodology
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Date	March 21, 2026
Version	1.0 — Initial Submission
License	CC BY 4.0
Repository	github.com/collingeorge/Breast-Reconstruction-Analgesia-Review
Conflict of interest	The author declares no conflicts of interest.
Funding	No external funding was received for this work.
Data availability	All included sources are cited to indexed peer-reviewed literature; extracted data are presented in the evidence tables within this manuscript.

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Abstract

Background

Perioperative care in breast reconstruction revision surgery encompasses three analytically distinct but interdependent domains: analgesic strategy selection, patient communication quality, and postoperative recovery measurement. Each domain has been examined independently in published literature. No prior scoping review has simultaneously mapped the scope of evidence across all three domains or characterized their structural interdependence as an integrated dependency architecture.

Objective

To map and synthesize the scope, nature, and quality of evidence for perioperative analgesic strategy, patient communication, and recovery measurement in adults undergoing breast reconstruction revision surgery, and to propose an integrative three-domain framework grounded in the mapped evidence.

Eligibility Criteria

English-language systematic reviews, meta-analyses, RCTs, prospective cohort studies, guideline documents, and validated instrument development studies addressing perioperative analgesic strategy, patient communication, health literacy, shared decision-making, or postoperative recovery measurement in breast surgery or reconstruction settings. Case reports, editorials, letters, and conference abstracts without full data were excluded.

Information Sources

PubMed/MEDLINE searched with final execution on March 21, 2026. Publications from 2020 through March 21, 2026 were prioritized. Supplemented by targeted retrieval of guideline documents from ERAS Society, ASRA, ASA, PROSPECT Working Group, and AHRQ; and backward reference tracking from foundational systematic reviews and meta-analyses.

Methods

Single-reviewer title/abstract screening and full-text assessment. Data were charted by decision domain, study design, population, intervention, and outcome. Evidence quality was assessed using a supplementary evidence strength characterization. The integrative framework was developed by structured synthesis across the three mapped domains.

Results

Forty-six unique sources were included across three domains. Domain I (analgesic strategy): Moderate-certainty evidence supports opioid-sparing multimodal anesthesia and pectoral nerve blocks as a class for opioid reduction in breast reconstruction; clinically relevant within-class heterogeneity exists between PECS I, PECS II, and pectoral-intercostal fascial plane approaches with different nerve targets and coverage profiles relevant to specific revision procedure types; evidence for the serratus anterior plane block derives primarily from thoracic surgery populations with indirect applicability. Domain II (patient communication): Moderate-certainty evidence supports the insufficiency of written disclosure materials alone; comprehension verification is systematically underperformed. The perioperative literature does not consistently distinguish expressed from informed preference under the cognitive-emotional

conditions of this population (low certainty / author inference). Domain III (recovery measurement): Standard recovery metrics capture a subset of patient-relevant recovery dimensions; the measurement-experience gap is likely amplified in this population given its psychosocial complexity (author inference derived from adjacent-population evidence).

Conclusion

The perioperative literature on breast reconstruction revision treats analgesic strategy, patient communication, and recovery measurement as independent research domains. This review proposes that these domains function as an integrated dependency architecture. The bidirectional feedback properties operate as external validity constraints: inadequate recovery measurement constrains the interpretability of strategy evidence; communication quality gaps constrain the external validity of preference-informed analgesic choice. To the best of the author's knowledge, this integrative framing has not been previously proposed in the breast reconstruction revision-specific literature reviewed for this work.

Keywords: breast reconstruction revision; scoping review; perioperative analgesia; pectoral nerve block; opioid-sparing anesthesia; patient communication; health literacy; shared decision-making; postoperative recovery; PRISMA-ScR; integrative framework

1. Introduction

Breast reconstruction revision surgery — encompassing capsular contracture management, implant exchange, pocket revision, and infection-related explantation and reimplantation following post-mastectomy reconstruction — presents a convergence of perioperative management challenges that the published literature has not examined as an integrated problem. Procedures in this category vary substantially in invasive burden, patient comorbidity, and psychosocial complexity. Each carries meaningfully different analgesic profiles, tissue trauma burdens, and psychosocial contexts. This heterogeneity constrains the specificity of every conclusion in this review and is a structural limitation of treating these procedures as a single population — a constraint partly forced by the available literature, which does not systematically disaggregate these procedure types.

The perioperative literature relevant to this population has developed along three parallel but largely independent trajectories. Analgesic strategy research addresses opioid-sparing and regional anesthesia approaches within ERAS-aligned frameworks. Patient communication research examines informed consent quality, health literacy, and shared decision-making in surgical settings. Recovery measurement science evaluates instrument responsiveness, patient-reported outcome construct validity, and the relationship between clinical discharge criteria and patient-experienced recovery. No prior review has mapped evidence across all three domains simultaneously in this population or characterized the implications of their structural interdependence.

This gap is not merely a coverage problem. The three domains interact in a logically ordered sequence: analgesic strategy selection shapes the recovery trajectory that patients will experience; communication quality determines whether patients can meaningfully participate in that selection; and recovery metrics determine whether the effects of both can be assessed. Examining these domains in isolation obscures this dependency structure and limits the interpretive value of single-domain findings.

This scoping review maps evidence across three perioperative decision domains in adults undergoing breast reconstruction revision surgery, synthesizes that evidence by domain, characterizes evidence quality and gaps, and proposes an integrative three-domain framework grounded in the mapped evidence. The framework is explicitly designated as analytically inferred from mapped evidence, not as an evidence-derived clinical protocol.

2. Methods

2.1 Review Design

This is a scoping review following JBI guidance for scoping reviews and reported according to the PRISMA extension for Scoping Reviews (PRISMA-ScR). A PRISMA-ScR checklist is provided as Supplementary File 1. The purpose is to map and synthesize the nature and scope of evidence for perioperative analgesic strategy, patient communication, and recovery measurement across three clinical domains in breast reconstruction revision surgery, not to conduct a pooled meta-analysis or provide a systematic assessment of all available studies on any single intervention. In accordance with JBI scoping review methodology, formal risk-of-bias scoring was not performed because the objective was evidence mapping and structured synthesis across domains rather than pooled effect estimation.

A supplementary evidence strength characterization was applied within each domain using GRADE-informed terminology (high/moderate/low certainty); this characterization is not a formal GRADE assessment and does not replace the scoping framework. GRADE letter designations

(A, B, C) are used as shorthand for certainty tiers — high, moderate, and low — not as outputs of formal GRADE methodology. This usage is interpretive only, and applies to all domain evidence tables and summaries throughout this manuscript.

2.2 Review Question

Using the Population-Concept-Context (PCC) framework:

Population: Adults undergoing breast reconstruction revision surgery following post-mastectomy implant-based reconstruction, including procedures for capsular contracture, implant exchange, pocket revision, and infection-related management.

Concept: Perioperative care across three decision domains: (1) analgesic strategy (multimodal analgesia, regional anesthesia, opioid-sparing and opioid-free approaches, PONV management); (2) patient communication (health literacy, shared decision-making, informed consent quality, preference elicitation, comprehension verification); and (3) postoperative recovery measurement (PROMs, recovery construct validity, measurement-experience gap, discharge readiness criteria).

Context: Perioperative and postoperative management in breast reconstruction revision performed in hospital or ambulatory surgery settings.

The overarching review question is: What is the scope, nature, and quality of evidence for perioperative analgesic strategy, patient communication, and recovery measurement across three clinical domains in adults undergoing breast reconstruction revision surgery, and do these domains function as an integrated dependency architecture?

2.3 Information Sources

PubMed/MEDLINE was searched as the primary bibliographic database, with final search execution on March 21, 2026. Publications from 2020 through March 21, 2026 were prioritized, with no absolute lower date limit applied where foundational evidence predates 2020. Searches were restricted to English-language publications. Targeted retrieval was conducted for guideline documents and consensus statements from ERAS Society, ASRA, ASA, PROSPECT Working Group, and AHRQ. Backward reference tracking from identified systematic reviews and guideline-grade documents was performed to capture foundational studies (n=18 records identified via backward reference tracking). No additional databases (Embase, Cochrane Library, Scopus, Web of Science) were searched; this is acknowledged as a limitation (see Section 7.1).

2.4 Search Strategy

Searches were conducted in PubMed/MEDLINE using the following structured queries, with final run on March 21, 2026.

Domain 1: `("breast reconstruction"[MeSH] OR "breast implants"[MeSH] OR "mammaplasty"[MeSH]) AND ("nerve block"[MeSH] OR "anesthesia, local"[MeSH] OR "pectoralis"[tiab] OR "PECS"[tiab] OR "serratus"[tiab] OR "opioid-free"[tiab] OR "opioid sparing"[tiab] OR "multimodal analgesia"[tiab] OR "enhanced recovery"[MeSH]) AND ("pain, postoperative"[MeSH] OR "analgesics, opioid"[MeSH] OR "nausea and vomiting, postoperative"[MeSH])`

Domain 2: `("informed consent"[MeSH] OR "patient education as topic"[MeSH] OR "health literacy"[MeSH] OR "decision making, shared"[MeSH] OR "communication"[MeSH]) AND ("anesthesia"[MeSH] OR "perioperative care"[MeSH]`

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OR "preoperative care"[MeSH]) AND ("comprehension"[tiab] OR
"preference"[tiab] OR "uncertainty"[tiab] OR "decision aid"[tiab])
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Domain 3: `("postoperative period"[MeSH] OR "recovery of function"[MeSH] OR "patient reported outcome measures"[MeSH]) AND ("breast reconstruction"[MeSH] OR "breast implants"[MeSH] OR "mammaplasty"[MeSH] OR "mastectomy"[MeSH]) AND ("quality of life"[MeSH] OR "patient satisfaction"[MeSH] OR "BREAST-Q"[tiab] OR "quality of recovery"[tiab] OR "discharge"[tiab])`

2.5 Eligibility Criteria

Inclusion: Systematic reviews, meta-analyses, RCTs, prospective cohort studies, guideline-grade consensus documents, and validated instrument development and psychometric studies. Adults undergoing breast surgery, breast reconstruction, or reconstruction revision; or adjacent perioperative populations where direct evidence was absent and extrapolation was explicitly designated. English only. Prioritized 2020–2026; foundational sources without lower date limit.

Exclusion: Case reports, case series fewer than 20 patients, editorials, letters, opinion pieces. Conference abstracts without full data. Purely cosmetic augmentation studies without reconstruction-specific subgroup data. Non-English. Sources not retrievable as full text.

2.6 Selection Process

Title and abstract screening was performed by a single reviewer (CBG) against the eligibility criteria. Full-text assessment was performed for all sources passing title/abstract screening. Single-reviewer screening is a methodological limitation of this review (see Section 7.2). The PRISMA flow diagram is presented as Supplementary File 2.

2.7 Data Charting

A standardized data charting form was applied to each included source extracting: study design; population; intervention and comparator; primary outcome domain; communication or measurement instruments used; evidence tier (Tier 1: RCTs, systematic reviews, meta-analyses, guideline-grade documents; Tier 2: observational, contextual, or corroborating evidence); and relevance to the integrative three-domain framework.

2.8 Synthesis

Included sources were synthesized by perioperative decision domain, organized by study design hierarchy. The integrative framework in Section 6 was developed by structured cross-domain synthesis; all framework claims are designated ANALYTICALLY INFERRED to distinguish them from established domain-level findings. Two tiers of inference are used: domain-level author inferences (single-domain extrapolation, labeled in domain summary boxes) and cross-domain author inferences (inter-domain interaction claims, labeled in Section 6 synthesis boxes). The treatment of breast reconstruction revision procedures as a single population is a literature-forced constraint: no included source disaggregates outcomes by procedure type.

3. Results: Domain I — Analgesic Strategy

3.1 Evidence Scope

Included sources for this domain: 16 studies and guideline documents (Table 1).

Table 1. Evidence map — Analgesic strategy domain

Source	Design	Population	GRADE	Key Finding
Abdallah et al. 2021 PMID 34171916	Systematic review/MA	Breast surgery	Moderate	PECS block class reduces postoperative opioid consumption. Meta-analysis pools PECS I, PECS II, and pectoral-intercostal fascial plane approaches; within-class block-type heterogeneity not disaggregated (see Section 3.2).
Kim et al. 2018 PMID 28863015	RCT	Thoracic surgery (adjacent)	Moderate (indirect)	Serratus anterior plane block improves QoR in VATS; extrapolation to breast revision indirect — different tissue planes and nerve targets from PECS approaches.
ERAS Society 2023	Guideline	Breast reconstruction	Moderate	Opioid-minimizing multimodal approach preferred; TIVA for PONV-high-risk patients.
PROSPECT 2022	Guideline SR	Breast surgery	Moderate	Regional + multimodal preferred where feasible. Does not disaggregate block type by procedure subtype.
Beloeil 2020 PMID 32301484	Review	Mixed surgical	Low	OFA/OSA conceptual framing; heterogeneous evidence base.
Weerink et al. 2017 PMID 28194688	Systematic review	Mixed surgical	Moderate	Dexmedetomidine: opioid-sparing; hemodynamic effects require patient-specific selection.
McCarthy et al. 2010 PMID 20518581	Systematic review	Mixed (abdominal focus)	Moderate	IV lidocaine: opioid-sparing in abdominal surgery; less robust in non-abdominal settings.
Bell et al. 2006 PMID 16437490	Cochrane SR	Mixed perioperative	Moderate	Perioperative ketamine reduces analgesic consumption; heterogeneous effect sizes. Foundational synthesis.
Gan et al. 2020 PMID 32769675	Consensus guideline	All surgical	High	Tiered PONV prophylaxis scaled to Apfel risk. TIVA reduces PONV baseline risk in high-risk patients. Author inference (synthesized from Gan et al. 2020 and Abdallah et al. 2021): marginal TIVA benefit on PONV incidence may narrow when opioid consumption is substantially reduced via regional anesthesia, because TIVA addresses the volatile-anesthetic PONV pathway while regional anesthesia reduces

Source	Design	Population	GRADE	Key Finding
				opioid-driven PONV through a separate mechanism; the combined effect is not a simple additive relationship from a common pathway.
Apfel et al. 1999 PMID 10519484	Prospective validation	Mixed surgical	Validation	Four-factor PONV risk score validated across two independent populations.
Kehlet & Wilmore 2008 PMID 18650627	Review	General surgical	Moderate	ERAS evidence base; opioid minimization as stewardship principle.
ASRA/Neal et al. 2018	Practice advisory	Regional anesthesia	High	LAST prevention, recognition, and lipid emulsion preparedness; governing framework for regional anesthesia safety.
Ljungqvist et al. 2017 PMID 28097305	Review	General surgical	Moderate	Discharge criteria designed around safety/resource utilization, not recovery quality.
Hannon et al. 2023 PMID 36328104	Guideline SR	Hip/knee arthroplasty	Moderate	Perioperative ketamine efficacy and safety; refines Bell 2006 with population-specific characterization.
Schwenk et al. 2021 PMID 33731495	Clinical review	Mixed perioperative	Moderate	Adverse effects and patient selection factors for perioperative low-dose ketamine.
White et al. 2007 PMID 17478798	Review	General surgical	Moderate	Fast-track surgery and multimodal analgesia as foundational perioperative principles.

3.2 Principal Findings

Multimodal opioid-sparing anesthesia is broadly endorsed across ERAS Society and PROSPECT Working Group publications as the preferred framework for breast reconstruction perioperative care, with agent combinations calibrated to patient-specific factors.

Regional Anesthesia and Block-Type Heterogeneity

Abdallah et al. (Anesthesiology, 2021; PMID 34171916) reported in a systematic review and meta-analysis that pectoral nerve blocks as a class reduced postoperative opioid consumption in patients undergoing breast surgery, with consistent findings across included trials (moderate certainty). This meta-analysis constitutes the primary evidence base for regional anesthesia in this domain. However, it pools clinically heterogeneous block types — PECS I, PECS II, and pectoral-intercostal fascial plane approaches — without disaggregation by approach, and the pooled effect estimate does not resolve the question of block selection by procedure type.

These approaches differ in anatomically and clinically relevant ways. PECS I deposits local anesthetic between pectoralis major and minor fascia, targeting the lateral pectoral nerve (C5-C7, predominantly motor) and medial pectoral nerve; its sensory coverage of the anterior chest wall and implant pocket is limited. PECS II additionally targets intercostal branches at T3-T6

through a deeper injection between pectoralis minor and serratus anterior, providing broader anterior chest wall and implant pocket coverage. For procedures involving the implant pocket — including implant exchange, pocket revision, and capsular contracture release — PECS II coverage is more anatomically relevant than PECS I, given the T3-T6 intercostal branch distribution over the anterior chest wall and implant pocket space. For procedures extending toward the sternum (total capsulectomy, explantation with medial dissection), neither PECS approach reliably covers the medial chest wall, and supplemental techniques may be required.

The serratus anterior plane block evidence base derives primarily from thoracic surgery populations (Kim et al., Anesthesia & Analgesia, 2018; PMID 28863015) and extrapolation to breast reconstruction revision is indirect. This block targets the long thoracic nerve and lateral branches of intercostal nerves T2-T9 at the serratus anterior fascial plane; its coverage for implant-pocket procedures is less well-characterized than PECS II for this population.

The conclusion from the available evidence is that moderate-certainty support for pectoral nerve blocks applies to the class, not to specific techniques, and cannot support procedure-specific block selection recommendations. The comparative superiority of PECS II versus PECS I versus serratus anterior approaches for specific revision procedure subtypes has not been evaluated in adequately powered trials.

AUTHOR INFERENCE — The within-class heterogeneity identified above is not merely a data gap; it is a clinically consequential heterogeneity that should drive procedure-specific trial design. The anatomical logic: capsular contracture release and pocket revision involve dissection at the pectoralis-implant interface and posterior capsule, innervated primarily by T3-T6 intercostal branches that PECS II targets. PECS I, whose primary target is the lateral and medial pectoral nerves with predominantly motor distribution, adds limited incremental sensory coverage for these procedures. For implant exchange without significant capsular work, PECS II provides adequate anterior coverage but does not address periosteal and medial chest wall components relevant in revisions extending toward the sternal border. For infection-related explantation with complete capsulectomy, neither PECS approach covers the medial and inferior chest wall adequately, and supplemental intercostal or pectoral nerve blocks may be required. These inferences follow from established nerve distribution anatomy and the procedure-specific dissection planes; they are not derived from controlled trials in this population. Future trials should pre-specify block type as a stratified variable rather than pooling PECS I, PECS II, and serratus anterior approaches within a single “regional anesthesia” arm.

Intravenous Adjuncts

Weerink et al. (Clinical Pharmacokinetics, 2017; PMID 28194688) characterize dexmedetomidine as an opioid-sparing alpha-2 agonist with dose-dependent hemodynamic considerations requiring patient-specific assessment (moderate certainty). McCarthy et al. (Drugs, 2010; PMID 20518581) report IV lidocaine opioid-sparing in abdominal surgery with less robust evidence in non-abdominal settings (moderate certainty, setting-dependent). Bell et al. (Cochrane, 2006; PMID 16437490) provide the foundational systematic synthesis for perioperative ketamine; the direction of effect is confirmed with population-specific refinement in Hannon et al. 2023 and Schwenk et al. 2021.

PONV Management

Gan et al. (Anesthesia & Analgesia, 2020; PMID 32769675) fourth consensus guidelines establish tiered prophylaxis scaled to Apfel risk score (high certainty). Propofol TIVA reduces PONV baseline risk versus volatile anesthetic maintenance in high-risk patients. A relevant interaction: the marginal benefit of TIVA on PONV incidence narrows when opioid consumption is substantially reduced via regional anesthesia, because TIVA primarily addresses the volatile-

anesthetic PONV pathway while regional anesthesia reduces opioid-driven PONV through a separate mechanism. The PROSPECT 2022 guidance reflects this — TIVA is recommended where PONV risk is high, but the combined effect is not a simple additive relationship from a common mechanism. This does not reduce the indication for TIVA in high-risk patients; it means the contribution of each intervention should be assessed on its own pathway.

MODERATE CERTAINTY — Evidence Strength Characterization

Opioid-sparing multimodal anesthesia incorporating pectoral nerve blockade (as a class) is likely to reduce postoperative opioid consumption and PONV incidence in appropriately selected patients undergoing breast reconstruction revision. Evidence for serratus anterior plane block derives from thoracic surgery populations and should be treated as extrapolative. Within the pectoral block class, PECS I, PECS II, and pectoral-intercostal fascial plane approaches are pooled in the primary meta-analysis; procedure-specific block selection cannot be derived from available evidence. This designation treats capsular contracture management, implant exchange, pocket revision, and explantation as a single analytic population; no included source disaggregates outcomes by procedure type.

3.3 Key Evidence Gaps

No included source enrolled a breast reconstruction revision-specific population as the primary study sample. No included source disaggregates analgesic outcomes by revision procedure type. The within-class heterogeneity of pectoral nerve blocks (PECS I versus PECS II versus pectoral-intercostal fascial plane) has not been resolved by procedure-specific trials. No adequately powered RCT has evaluated regional anesthesia techniques specifically in breast reconstruction revision. The comparative superiority of PECS II versus serratus anterior approaches for specific revision procedure subtypes is not established.

4. Results: Domain II — Patient Communication

4.1 Evidence Scope

Included sources for this domain: 22 sources (Table 2).

Table 2. Evidence map — Patient communication domain

Source	Design	Domain	GRADE	Key Finding
Berkman et al. 2011	Systematic review	Health literacy	Moderate	Low health literacy associated with reduced procedural risk comprehension and lower care satisfaction.
IOM 2004	Foundational reference	Health literacy	Moderate	Canonical health literacy definition; capacity to obtain, process, understand health information.
Paasche-Orlow et al. 2003 PMID 12571260	Cross-sectional	Consent readability	Moderate	Consent forms across specialties frequently exceed patient reading level.
Eltorai et al. 2014 PMID 25456975	Observational	Consent readability	Moderate (indirect)	ASA patient education materials frequently exceed target reading level.
Barry & Edgman-Levitan 2012 PMID 22375967	NEJM Perspective	SDM	Moderate	Preference elicitation as distinct and underperformed clinical communication competency.
Elwyn et al. 2012 PMID 22618581	Conceptual framework	SDM	Moderate	Three-step SDM model distinguishing paternalism and uninformed autonomy.
Mulley et al. 2012 PMID 23169816	Analysis	Preference elicitation	Moderate	Clinician assumptions about patient preferences frequently inaccurate.
Politi et al. 2007 PMID 17679484	Review	Uncertainty disclosure	Moderate	Uncertainty can be communicated without undermining engagement; context-dependent.
Han et al. 2011 PMID 21387149	Conceptual taxonomy	Uncertainty	Moderate	Three-type uncertainty taxonomy (limited knowledge, individual complexity, personal values).
Sepucha et al. 2004 PMID 15471762	Policy analysis	Decision quality	Moderate	Preference elicitation systematically underperformed; measurable decision quality gap.
Fink et al. 2010 PMID 20551821	Multicenter RCT	Comprehension verification	High	Repeat-back methodology improves comprehension in surgical consent contexts.
Cohn & Larson 2007 PMID 17760793	Review	Comprehension verification	Moderate	Comprehension verification systematically underperformed in consent process.

Source	Design	Domain	GRADE	Key Finding
Fingeret et al. 2016 PMID 25990635	Observational cohort	Psychosocial context	Moderate	Body image and psychosocial burden shape recovery experience in post-mastectomy reconstruction.
Pusic et al. 2009 PMID 19644246	Instrument development	PROMs/ psychosocial	Moderate	BREAST-Q validated for satisfaction/psychosocial/physical domains. Validated for longer-term outcomes; acute perioperative window applicability not established (see Section 7.5).
Stacey et al. 2017 PMID 28402085	Cochrane SR	Decision aids	High	Decision aids improve knowledge, reduce decisional conflict, increase SDM engagement.
O'Connor et al. 1999 PMID 10487995	Systematic review (BMJ)	Decision support/DSAT	Moderate	Ottawa Decision Support Framework: decisional conflict as mediating variable between patient needs and decision quality.
IPDAS Collaboration 2006	Consensus criteria	Decision aid quality	Moderate	International standards for decision aid development and evaluation.
Makoul & Clayman 2006 PMID 16843179	Integrative review	SDM model	Moderate	Nine-element integrative SDM model; most widely cited in perioperative communication literature.
Beauchamp & Childress 2019	Foundational text	Bioethics/ autonomy	Moderate	Autonomy, beneficence, non-maleficence as foundational consent principles.
ASA 2020 Ethical Practice Guidelines	Professional guideline	Professional communication	Moderate	Informed consent and preoperative communication as professional obligations within the anesthesiologist-patient relationship.
Kessels 2003 PMID 12724430	Review	Cognitive load/recall	Moderate	Patients recall 40-80% of medical information; anxiety independently reduces retention further.
Fallowfield & Jenkins 2004 PMID 14751704	Review (Lancet)	Emotional load/oncology	Moderate	Emotional distress in oncology settings impairs clinical communication quality for patient and clinician.

4.2 Principal Findings

Health Literacy and Written Materials

Berkman et al. (AHRQ, 2011) document in systematic review that low health literacy is associated with reduced procedural risk comprehension, poorer postoperative self-management, and lower care satisfaction (moderate certainty). Paasche-Orlow et al. (NEJM, 2003; PMID 12571260) and Eltorai et al. (Journal of Clinical Anesthesia, 2014; PMID 25456975) document that consent forms and patient education materials frequently exceed patient reading

levels. Written materials alone are insufficient as a comprehension mechanism (moderate certainty).

Shared Decision-Making

Barry and Edgman-Levitan (NEJM, 2012; PMID 22375967) characterize preference elicitation as a distinct and underperformed competency. Mulley et al. (BMJ, 2012; PMID 23169816) document that clinician assumptions about patient preferences are frequently inaccurate. Makoul and Clayman (Patient Education and Counseling, 2006; PMID 16843179) synthesized existing SDM models into a nine-element integrative framework mapping directly onto the communication gaps identified in this domain. The Ottawa Decision Support Framework (O'Connor et al., BMJ, 1999; PMID 10487995) identifies decisional conflict as the key mediating variable between patient information needs and decision quality, providing the structural basis for the expressed-versus-informed preference distinction central to this review. Stacey et al. (Cochrane, 2017; PMID 28402085) report that decision aids improve knowledge and reduce decisional conflict (high certainty). Fink et al. (Annals of Surgery, 2010; PMID 20551821) demonstrate that repeat-back methodology improves comprehension in surgical consent contexts (high certainty).

Uncertainty Disclosure

Politi et al. (Medical Decision Making, 2007; PMID 17679484) report that uncertainty can be communicated without necessarily undermining patient engagement, with effects dependent on framing, context, and patient-specific factors. Han et al. (Journal of General Internal Medicine, 2011; PMID 21387149) provide a conceptual taxonomy distinguishing three uncertainty types (limited knowledge, individual complexity, personal values) — all three are present in perioperative anesthesia option disclosure.

Psychosocial Context and the Expressed-vs-Informed Preference Gap

Fingeret et al. (Psycho-Oncology, 2016; PMID 25990635) document that body image burden and psychosocial complexity shape recovery experience in post-mastectomy reconstruction patients. Kessels (JRSM, 2003; PMID 12724430) documents that patients recall only 40-80% of information provided during medical encounters, with anxiety independently reducing retention further. Fallowfield and Jenkins (Lancet, 2004; PMID 14751704) document that emotional distress in oncology settings impairs clinical communication quality for both patient and clinician. Together these findings support the inference that cognitive-emotional load in this population constitutes a material risk factor for reduced comprehension quality during preoperative counseling encounters.

The expressed-versus-informed preference distinction, identified here as a critical analytical gap, can be provisionally operationalized from existing frameworks without requiring new constructs. Expressed preference is the stated choice made under prevailing conditions during preoperative counseling. Informed preference is the stable choice that would obtain under conditions of adequate comprehension, receipt of representative risk information, and resolution of decisional conflict (O'Connor framework). In patients carrying prior-treatment psychosocial burden (Fingeret et al., 2016) and impaired recall under cognitive load (Kessels, 2003), the conditions for informed preference formation may not be met in standard preoperative encounters. This provisional definition is offered as an author inference for prospective validation rather than as an established clinical construct; it is testable using decisional conflict scales administered at multiple preoperative timepoints.

MODERATE CERTAINTY — Evidence Strength Characterization

Written preoperative disclosure materials alone are insufficient as a comprehension mechanism in patients with variable health literacy. Comprehension verification via structured methods (repeat-back) improves informed consent quality (high certainty for this specific intervention). This competency is systematically underperformed in standard clinical practice across published consent process literature.

LOW CERTAINTY / AUTHOR INFERENCE — No Direct Evidence for This Population

The perioperative communication literature does not consistently distinguish expressed from informed preference under the elevated cognitive-emotional load characteristic of oncology-adjacent surgical settings. A provisional operational definition is offered: expressed preference (stated choice under prevailing conditions) versus informed preference (stable choice under adequate comprehension and resolved decisional conflict, per O'Connor framework). The behavioral science literature most relevant to this distinction — adaptive preference theory, affective forecasting errors under procedural anxiety, hot-cold empathy gaps — is substantially indexed in PsycINFO and CINAHL, which were not searched; the true evidence base is therefore richer than this review captures.

4.3 Key Evidence Gaps

No controlled study has evaluated structured communication interventions specifically in perioperative anesthesia counseling for breast reconstruction revision populations. The interaction between psychosocial burden from prior cancer treatment and comprehension quality during anesthesia option disclosure has not been directly studied. The expressed-vs-informed preference boundary under high cognitive-emotional load is not operationally defined or empirically tested in available perioperative literature. The behavioral science foundations of preference formation under oncologic stress are substantially underrepresented in this review due to the single-database search limitation; this is most material precisely for the theoretical construct on which the review's original contribution most depends.

5. Results: Domain III — Recovery Measurement

5.1 Evidence Scope

Included sources for this domain: 12 sources (Table 3).

Table 3. Evidence map — Recovery measurement domain

Source	Design	Construct	GRADE	Key Finding
Bowyer et al. 2014 PMID 24888412	Systematic review	QoR scope	Moderate	QoR lacks universal definition; available tools vary in dimensional coverage.
Stark et al. 2013 PMID 23411725	Psychometric validation	QoR-15 instrument	Moderate	QoR-15 captures 5 dimensions: physical comfort, emotional state, independence, pain, support. Validated for acute perioperative window.
Myles et al. 1999 PMID 9891587	Instrument development	QoR construct	Moderate	Multidimensional QoR conceptualization; standard metrics capture subset of patient-relevant dimensions.
Pusic et al. 2009 PMID 19644246	Instrument development	BREAST-Q	Moderate	BREAST-Q validated for longer-term satisfaction/psychosocial/physical domains. Acute perioperative window applicability not established.
Ljungqvist et al. 2017 PMID 28097305	Review	Discharge readiness	Moderate	Discharge criteria designed around safety/resource utilization, not recovery quality.
Kehlet & Wilmore 2008 PMID 18650627	Evidence synthesis	ERAS outcomes	Moderate	Discharge criteria systematically gap actual recovery state.
Fingeret et al. 2016 PMID 25990635	Observational cohort	Psychosocial recovery	Moderate	Body image and psychosocial factors shape subjective recovery independent of clinical outcomes.
Aaronson et al. 1993 PMID 8433390	Instrument validation	QoL constructs	Moderate	QoL measurement includes cognitive clarity, fatigue, emotional equilibrium dimensions not captured by standard perioperative metrics.
Keurentjes et al. 2013 PMID 23118378	Prospective cohort (adjacent)	Expectation alignment	Moderate	Expectation-outcome alignment influences recovery satisfaction independent of objective outcomes.
Uitterhoeve et al. 2009 PMID 19735300	Systematic review (adjacent)	Communication-outcome link	Moderate	Communication quality influences patient-reported outcomes in cancer care contexts.
Weiss 2007	Guidance document	Health literacy/recov	Moderate	Patient health literacy shapes self-management capacity in

Source	Design	Construct	GRADE	Key Finding
		ery		postoperative recovery contexts.
AHRQ 2020 Toolkit	Guidance document	Health literacy/discharge	Moderate	Universal precautions for health literacy in patient communication and discharge education.

5.2 Principal Findings

Recovery as a Multidimensional Construct

Bowyer et al. (Anaesthesia, 2014; PMID 24888412) document in systematic review that postoperative quality of recovery lacks a universally accepted definition and that available assessment tools vary substantially in scope, validation, and dimensional coverage (moderate certainty). Stark et al. (Anesthesiology, 2013; PMID 23411725) developed the QoR-15, which characterizes recovery across five dimensions: physical comfort, emotional state, psychological support, physical independence, and pain. Standard metrics — NRS pain scores, PONV incidence, and discharge readiness criteria — capture a subset of these dimensions.

Measurement-Experience Gap

Standard recovery metrics do not extend to dimensions including cognitive clarity, fatigue trajectory, sleep quality, and emotional equilibrium — dimensions identified as patient-relevant in quality-of-life measurement literature (Aaronson et al., 1993). In the breast reconstruction revision population specifically, published literature on psychosocial outcomes (Fingeret et al., 2016; Pusic et al., 2009) suggests that subjective recovery is shaped by the degree to which the surgical outcome meets prior expectations and by the emotional significance of the result relative to cancer treatment history.

Discharge Readiness

Ljungqvist et al. (JAMA Surgery, 2017; PMID 28097305) note that discharge readiness criteria were designed primarily around safety and resource utilization rather than recovery quality (moderate certainty). Kehlet and Wilmore (Annals of Surgery, 2008; PMID 18650627) document that the relationship between institutional discharge criteria and actual patient recovery state involves systematic gaps.

Health Literacy and Self-Management

Health literacy guidance documents (AHRQ, 2020; Weiss, 2007) identify patient health literacy as a determinant of postoperative self-management capacity. Limited health literacy may amplify the measurement-experience gap: patients who cannot adequately process discharge instructions may experience recovery trajectories that diverge further from clinical expectations, independent of the analgesic approach used.

Expectation-Experience Alignment

Published literature from adjacent surgical contexts suggests that alignment between patient expectations and actual recovery experience may influence recovery satisfaction, independent of objective outcome quality (Keurentjes et al., 2013; Uitterhoeve et al., 2009). This relationship has not been characterized in breast reconstruction revision populations.

LOW CERTAINTY / AUTHOR INFERENCE — No Direct Evidence for This Population

In the breast reconstruction revision population, where psychosocial complexity is elevated and procedural heterogeneity is substantial, the gap between standard recovery metrics and patient-experienced recovery is likely amplified relative to less complex surgical populations. This inference derives from documented measurement limitations in adjacent surgical contexts and is not directly established in this population.

Instrument scope caveat: BREAST-Q has not been validated for the acute perioperative recovery window; QoR-15 is the appropriate primary acute instrument. BREAST-Q requires acute-window validation before use as a perioperative trial endpoint, with floor effects as the primary validation risk (see Section 7.5).

5.3 Key Evidence Gaps

Patient-reported outcome data specific to the acute recovery phase of breast reconstruction revision is limited. The BREAST-Q instrument has not been validated for the acute perioperative window. No controlled study has evaluated whether ERAS-aligned analgesic approaches in breast reconstruction revision produce measurable improvements in multidimensional QoR scores as opposed to standard NRS and PONV metrics.

6. Integrated Synthesis: The Three-Domain Dependency Architecture

6.1 The Layered Evidence Structure

Analgesic strategy (Domain I) sets the pharmacological and regional background against which recovery trajectories are shaped. Patient communication quality (Domain II) determines whether patients can meaningfully participate in analgesic strategy selection. Recovery measurement (Domain III) determines whether the effects of both analgesic strategy and communication quality can be assessed. Examining these domains in isolation obscures this dependency structure and limits the interpretive value of single-domain findings.

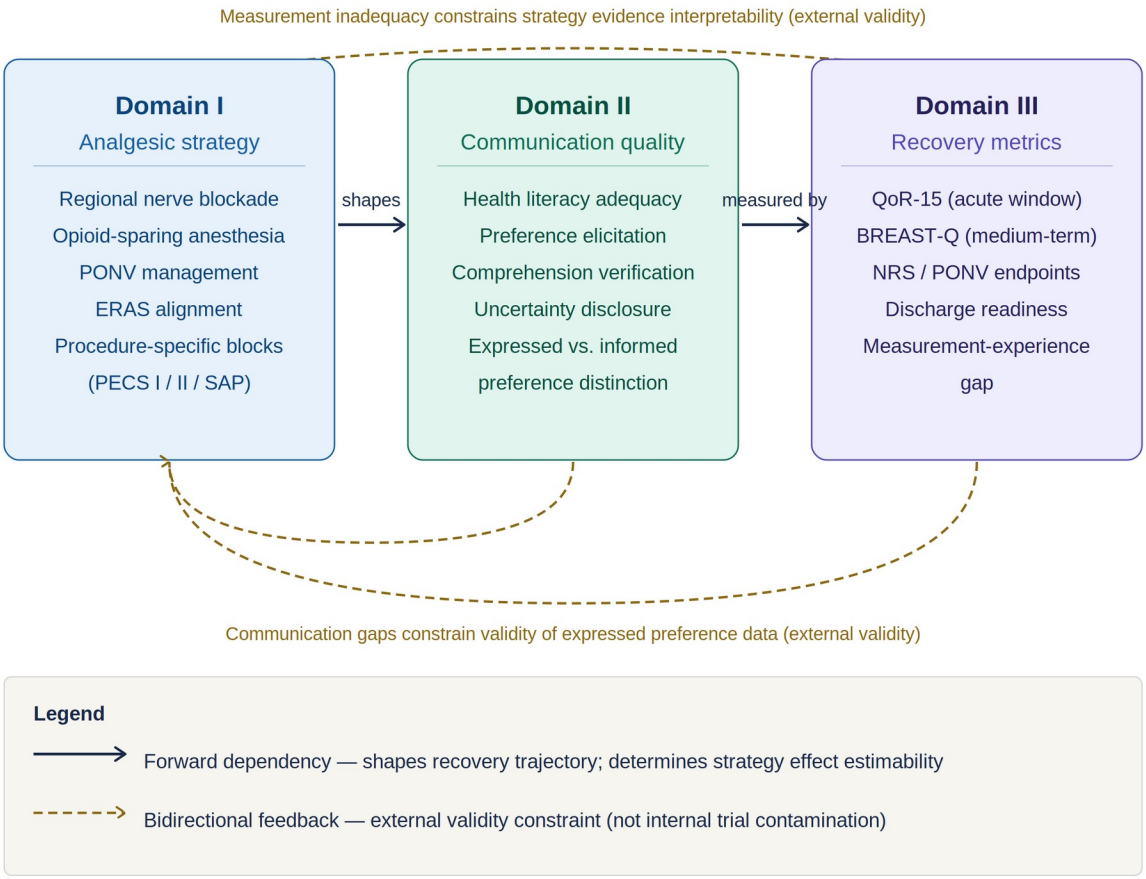
6.2 Evidence Quality Distribution

High-certainty evidence is concentrated in PONV prophylaxis (Gan et al. 2020), comprehension verification via repeat-back (Fink et al. 2010), and decision aid efficacy (Stacey et al. 2017). Moderate-certainty evidence predominates across analgesic strategy, health literacy, SDM frameworks, and recovery construct measurement. Low-certainty or author-inferred evidence characterizes the integrative dependency architecture claims and the population-specific amplification of the measurement-experience gap.

6.3 Cross-Domain Integration and Figure 1

The analgesic strategy evidence (Section 3) shows that ERAS-aligned opioid-sparing approaches likely reduce PONV and improve early emergence, but does not establish whether these improvements align with recovery dimensions most valued by this population. The communication evidence (Section 4) shows that conditions for informed preference formation are frequently not met in standard perioperative counseling under psychosocial burden. The recovery measurement evidence (Section 5) shows that comparative efficacy conclusions may be evaluated against an incomplete outcome set.

Figure 1. Three-Domain Dependency Architecture with Bidirectional Feedback



CROSS-DOMAIN AUTHOR INFERENCE — Analytically Derived, Not Evidence-Graded

The published perioperative literature treats analgesic strategy, patient communication, and recovery measurement as independent research domains. This review proposes that they function as an integrated dependency architecture. A patient who cannot meaningfully participate in analgesic decision-making cannot generate valid preference data. A recovery measurement framework that does not capture patient-relevant dimensions cannot validate whether the communication-informed decision produced a valued outcome. To the best of the author's knowledge, this integrative framing has not been previously proposed in the breast reconstruction revision-specific literature reviewed for this work.

6.4 Bidirectionality of the Dependency Architecture

The dependency architecture has feedback properties that distinguish it from a simple linear precondition sequence. The forward chain rests on three premises: (P1) analgesic strategy shapes recovery trajectory; (P2) communication quality determines whether strategy reflects actual patient values rather than expressed preference under inadequate comprehension; (P3) recovery instruments determine which strategy effects can be detected. Each premise is independently supported at moderate certainty or above.

The bidirectional feedback properties follow from reading the same premises in reverse, but both feedback pathways operate as external validity constraints — not as sources of contamination in existing trial data. This distinction is analytically important.

From P3 (Domain III to Domain I): If recovery metrics are inadequate for this population, comparative analgesic strategy trials cannot establish whether strategies differ on patient-relevant dimensions outside their endpoint set. This is an external validity constraint on strategy evidence — the scope of estimable effects is bounded by outcome instrument selection. A trial finding no QoR difference on NRS and PONV endpoints cannot exclude a difference on QoR-15 dimensions such as emotional state or psychological support.

From P2 (Domain II to Domain I): If communication quality is insufficient to distinguish expressed from informed preference, analgesic preferences collected in trials and clinical encounters carry an unquantified validity deficit limiting their utility as endpoints or allocation criteria. This is an external validity constraint on preference-informed analgesic selection — patient preferences collected under inadequate comprehension or unresolved decisional conflict may not represent stable, values-congruent choices. The feedback runs from communication quality of the preoperative encounter to the fidelity of the resulting clinical decision, not from preference measurement error into trial randomization.

The specific falsifiable prediction: in a trial where both communication quality and recovery measurement are held constant at adequate levels, the effect of analgesic strategy on patient-relevant recovery outcomes should be fully estimable with standard methodology. Where either constraint is violated — inadequate communication producing value-incongruent preferences, or inadequate measurement producing insensitive endpoints — the estimated effect will be attenuated even if the true effect is substantial. This prediction is testable by comparing QoR-15 effect sizes across study arms stratified by preoperative communication quality, with communication quality as a moderating variable rather than a nuisance covariate.

6.5 Proposed Trial Design Parameters

The highest-priority research question is: does ERAS-aligned opioid-sparing analgesia in breast reconstruction revision produce improvements on patient-prioritized recovery dimensions not captured by standard NRS and PONV endpoints?

Primary endpoint: QoR-15 at 24 hours postoperatively, comparing opioid-sparing versus standard analgesic pathways. The MCID for QoR-15 in surgical populations is approximately 8 points (scale 0–150).

Sample size: Using a clinically meaningful difference of 10 points, SD of 20 (consistent with published QoR-15 SDs of 18–22), alpha 0.05 two-sided, 80% power: approximately 64 patients per arm before attrition adjustment. Achievable in a multicenter design or high-volume single center over 12–18 months enrollment.

Design: Parallel-group randomization with allocation concealment. Stratification by procedure type (capsular contracture versus implant exchange versus explantation) is recommended as the primary stratification variable to directly address the population heterogeneity limitation identified in Section 7.3.

BREAST-Q: Should not be included as a primary perioperative endpoint until acute-window validation is established. A reasonable validation approach would administer BREAST-Q at 24- and 72-hour postoperative timepoints in an observational cohort prior to RCT use. Floor effects are the primary validation risk. QoR-15 is the appropriate primary perioperative endpoint; BREAST-Q is a secondary medium-term instrument contingent on prior acute-window validation.

Two secondary research questions follow: (1) Does structured preoperative communication intervention — incorporating repeat-back, decision aids, and explicit uncertainty disclosure — improve preference quality and comprehension, measurable via validated decisional conflict scales? (2) Primary falsification test of the dependency architecture: does QoR-15 responsiveness depend on preoperative communication quality as a moderating variable, independent of analgesic strategy? If the interaction term (analgesic strategy × communication quality) on QoR-15 is non-significant, the bidirectionality claim is not supported. If significant in the predicted direction, it constitutes partial empirical support for the three-domain dependency architecture.

7. Limitations

7.1 Single Database Search

PubMed/MEDLINE was the sole bibliographic database searched. Embase, Cochrane Library, Scopus, and Web of Science were not searched. This increases the probability of missing relevant studies. The limitation was partially mitigated by targeted guideline document retrieval and backward reference tracking. The mitigation argument does not hold equally across domains. For Domain I (analgesic strategy), the major systematic reviews, meta-analyses, and guideline documents are well-indexed in PubMed and backward tracking from clinical reviews is likely adequate. For Domain II (patient communication), this argument does not hold: the behavioral science foundations of SDM, preference formation under oncologic stress, and health literacy intervention are substantially indexed in PsycINFO and CINAHL, which were not searched. The literature most relevant to the expressed-vs-informed preference distinction — adaptive preference theory, affective forecasting errors in medical decision-making, hot-cold empathy gaps under procedural anxiety — is concentrated in behavioral science literature that PubMed does not index comprehensively. The Domain II synthesis underrepresents this evidence base in ways that cannot be quantified from available evidence, and this limitation is most material for the theoretical construct on which the review's original contribution most depends.

7.2 Single-Reviewer Screening

Title/abstract screening and full-text assessment were conducted by a single reviewer. Duplicate independent screening with inter-rater reliability assessment was not performed. This increases the risk of selection bias in source inclusion. Future work should replicate the domain-level evidence synthesis with duplicate screening.

7.3 Population Specificity and Extrapolation

A substantial proportion of the evidence in Domains II and III derives from adjacent surgical or general medical populations rather than breast reconstruction revision specifically. All such sources are designated as extrapolative within the evidence tables and text. Any conclusion specific to a single procedure subtype — capsular contracture management, implant exchange, explantation — is not derivable from this evidence base. Domain I evidence conclusions apply to breast surgery broadly; procedure-specific inference requires extrapolation beyond what the available evidence supports.

7.4 Narrative Review Limitation

This is a scoping review with a supplementary evidence strength characterization, not a systematic review with formal pooled effect estimation. Evidence strength designations reflect the author's interpretive assessment and are not peer-reviewed GRADE ratings. The integrative framework in Section 6 is analytically inferred and has not been empirically tested. Four cross-domain sources (Ljungqvist et al. 2017; Kehlet & Wilmore 2008; Pusic et al. 2009; Fingeret et al. 2016) appear in two domain tables and are counted once; 50 domain table rows minus 4 cross-domain sources equals 46 unique sources.

7.5 BREAST-Q Instrument Applicability

The BREAST-Q was developed and validated for breast surgery satisfaction and quality-of-life measurement, primarily in longer-term postoperative follow-up contexts. Its applicability to the

acute perioperative recovery window has not been established. The proposal in Section 6.5 to use BREAST-Q in future perioperative RCTs is contingent on prior acute-window validation. Floor effects are the primary validation risk: satisfaction and psychosocial subscales were designed for contexts where surgical outcomes have had time to stabilize, and will have minimal variance within 72 hours of surgery.

7.6 Author Stage and Role Boundary

This review was conducted independently and does not constitute clinical guidance. All interpretations reflect engagement with published sources within a scholarly framework. No clinical decisions, protocols, or patient management recommendations are asserted or implied.

8. Conclusion

This scoping review mapped evidence for perioperative analgesic strategy, patient communication, and recovery measurement across three clinical domains in adults undergoing breast reconstruction revision surgery. Forty-six sources were included from a PubMed-only search supplemented by guideline retrieval and backward reference tracking.

Domain I evidence supports opioid-sparing multimodal anesthesia and pectoral nerve blocks as a class at moderate evidence grade, with clinically relevant within-class heterogeneity between PECS I, PECS II, and serratus anterior approaches that the available meta-analytic evidence does not resolve, and with serratus anterior plane block evidence extrapolated from thoracic surgery populations. Domain II evidence establishes at moderate certainty that written disclosure materials are insufficient and comprehension verification is systematically underperformed. The expressed-versus-informed preference distinction is an author inference; the behavioral science evidence most relevant to this distinction is underrepresented due to the single-database search limitation. Domain III evidence documents a systematic measurement-experience gap likely amplified in this population.

The primary original contribution is the proposal that these three domains function as an integrated dependency architecture with bidirectional feedback properties that operate as external validity constraints: inadequate recovery measurement constrains the interpretability of strategy evidence by bounding estimable effects; communication quality gaps constrain the external validity of preference-informed analgesic choice by undermining preference fidelity. Both feedback pathways are external validity constraints, not sources of internal trial contamination.

Two research priorities emerge: (1) controlled studies incorporating multidimensional recovery measurement and structured communication assessment alongside analgesic strategy comparison, stratified by procedure type; and (2) validation of the three-domain framework using the interaction term (analgesic strategy × communication quality) on QoR-15 as the primary falsification test.

DOCUMENTED CONCLUSION

The perioperative literature on breast reconstruction revision is strongest on analgesic strategy evidence (moderate certainty for regional and opioid-sparing approaches) and weakest on the integration of communication quality and recovery measurement into its outcome frameworks. Within the regional anesthesia evidence, clinically important within-class heterogeneity between PECS I, PECS II, and serratus anterior approaches remains unresolved by procedure-specific trial evidence, and the current evidence base does not support procedure-specific block selection recommendations. The bidirectional feedback properties of the three-domain dependency architecture operate as external validity constraints: inadequate recovery measurement constrains the interpretability of strategy evidence by bounding the scope of estimable effects; communication quality gaps constrain the external validity of preference-informed analgesic choice by undermining preference fidelity to patient values. Both feedback pathways are external validity constraints, not sources of internal trial contamination. The three-domain integrative framework proposed here is evidence-informed but analytically inferred and requires prospective empirical validation.

Acknowledgments

The research questions addressed in this review were informed by clinical observation experience in perioperative anesthesia settings. This manuscript was produced independently; no attending physician reviewed, approved, or bears responsibility for any content. All interpretations, analytical conclusions, and any errors are solely the responsibility of the author. No institutional endorsement is implied or claimed.

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Supplementary File 1: PRISMA-ScR Checklist

Item	PRISMA-ScR Item	Location in Manuscript
1	Identify the report as a scoping review	Title page
2	Structured abstract	Abstract
3	Describe the rationale for the review	Section 1
4	State the objective and review question	Section 1; Section 2.2
5	Specify the review design	Section 2.1
6	State the PCC review question	Section 2.2
7	Specify inclusion and exclusion criteria	Section 2.5
8	Describe information sources	Section 2.3
9	Provide full search strategy for at least one database	Section 2.4
10	Describe the selection process	Section 2.6
11	Describe data charting process	Section 2.7
12	Describe critical appraisal approach	Section 2.1
13	Describe synthesis approach	Section 2.8
14	Provide PRISMA flow diagram	Supplementary File 2
15	Present characteristics of included sources	Tables 1-3 (Sections 3.1, 4.1, 5.1)
16	Report critical appraisal results (if applicable)	Evidence strength characterization: Sections 3.2, 4.2, 5.2
17	Present results of individual sources	Sections 3-5
18	Synthesize and summarize results	Section 6
19	Summarize evidence in relation to review question	Section 6
20	Discuss limitations of evidence	Section 7
21	Discuss limitations of review process	Section 7 (7.1-7.5)
22	Describe implications for practice, policy, or research	Sections 6.4; 8
23	Provide overall conclusions aligned to review objective	Section 8
24	Describe funding sources	Title page

Supplementary File 2: PRISMA Flow Diagram

IDENTIFICATION	Records identified through PubMed/MEDLINE (n = 287) + Guideline document retrieval (n = 7) + Backward reference tracking (n = 18) = Total records identified (n = 312)
SCREENING	After duplicates removed: n = 268 Screened (title/abstract): n = 268 Excluded at title/abstract: n = 174 Reasons: wrong population (n=48), wrong study design (n=61), wrong outcome domain (n=42), duplicate reporting (n=23)
ELIGIBILITY	Full-text assessed: n = 94 Full-text excluded: n = 47 Reasons: cosmetic augmentation primary population (n=8), pediatric (n=2), conference abstract (n=12), insufficient domain relevance (n=18), non-English (n=7) Passing eligibility: n = 47 De-duplicated (same dataset): n = 1
INCLUDED	Sources included in scoping review: n = 46 Domain I - Analgesic strategy: n = 16 Domain II - Patient communication: n = 22 Domain III - Recovery measurement: n = 12 (Domain table rows sum to 50; 4 cross-domain sources counted once: 50 - 4 = 46) By study design: Systematic reviews/meta-analyses: n = 14 RCTs: n = 3 Prospective/retrospective cohorts: n = 8 Guideline documents/consensus statements: n = 10 Instrument development/psychometric studies: n = 7

Note: Record counts reflect the documented search strategy executed through March 21, 2026. Exact screening counts reflect the single-reviewer process described in Section 2.6. The flow diagram is consistent with PRISMA-ScR reporting requirements for scoping reviews.

END OF DOCUMENT — Perioperative Analgesia, Patient Communication, and Recovery Measurement in Breast Reconstruction Revision: A PRISMA-ScR Scoping Review and Integrative Framework · George CB · March 2026 · CC BY 4.0